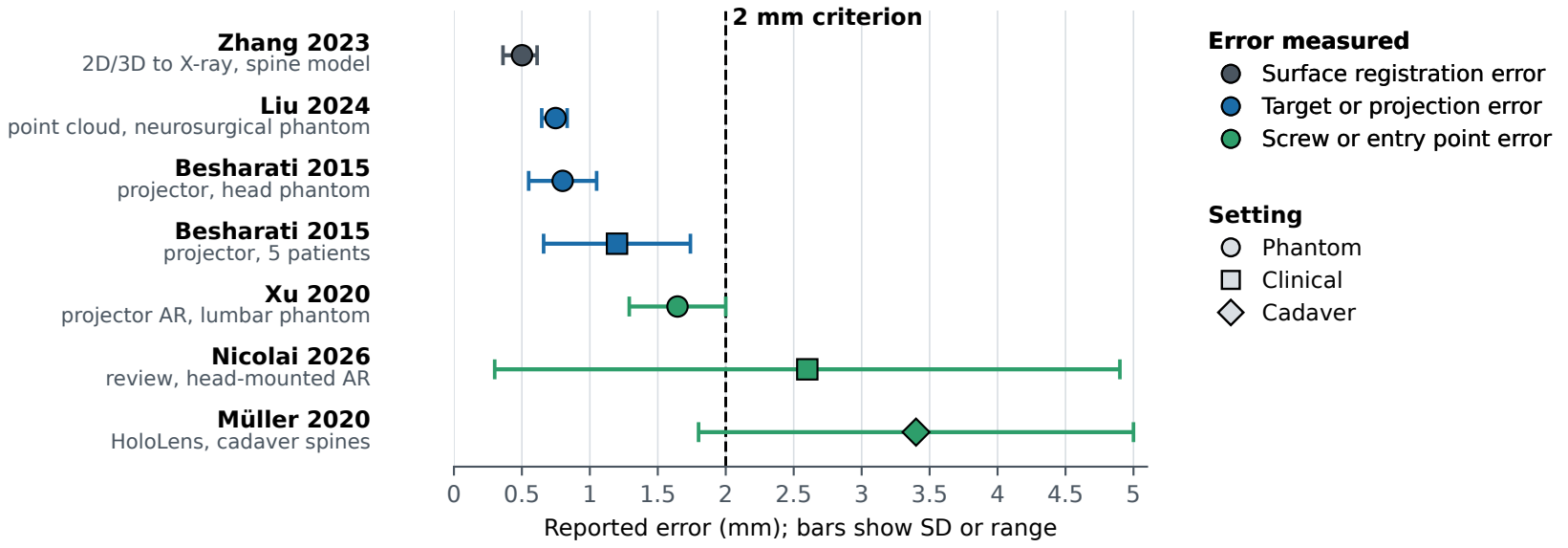
Zhang 2023
2D/3D to X-ray, spine model
Liu 2024
point cloud, neurosurgical phantom
Besharati 2015
projector, head phantom
Besharati 2015
projector, 5 patients
Xu 2020
projector AR, lumbar phantom
Nicolai 2026
review, head-mounted AR
Müller 2020
HoloLens, cadaver spines
2 mm criterion
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
Reported error (mm); bars show SD or range
Error measured
Surface registration error
Target or projection error
Screw or entry point error
Setting
Phantom
Clinical
Cadaver